

Protocol Flow

black-box SQR $\rightarrow$ calibrated displacement $D(\alpha)$ $\rightarrow$ known dispersive wait t

$\rightarrow$ qubit pre-rotation for X/Y/Z tomography $\rightarrow$ z readout $\rightarrow$ inverse fit on recoverable sector data

Key inference target: weighted transverse sector amplitudes $u_n = p_n (X_n + i Y_n)$

Key limitation: the allowed protocol does not separate p_n and Z_n .

Recommendation: Use it only when the goal is the cleanest recovery of the weighted transverse sector data.

Add displacement-plus-wait when you need a diagnostic for cavity coherences or model mismatch.

Do not claim recovery of p_n or sector-resolved Z_n without an extra cavity-sensitive measurement primitive.